Chemical and Mechanical Pathology of VRF System Failures: Moisture Contamination, Lubricant Degradation, and Copper Plating

The transition from traditional chlorofluorocarbon (CFC) and hydrochlorofluorocarbon (HCFC) refrigerants to modern hydrofluorocarbon (HFC) blends, most notably R410A, necessitated a fundamental and complete paradigm shift in refrigeration tribology. Because HFC refrigerants lack the chlorine molecules that previously provided natural boundary lubricity in legacy systems, and because they are completely immiscible with traditional non-polar mineral oils and alkylbenzenes, the HVAC/R industry was forced to adopt highly polar synthetic lubricants to ensure proper oil return through the complex piping networks of Variable Refrigerant Flow (VRF) systems.[1] This chemical polarity, while absolutely essential for refrigerant miscibility, introduced a severe and inescapable vulnerability into the installation process: extreme hygroscopicity. Synthetic oils aggressively attract and bond with moisture at a molecular level, fundamentally changing the physics of system evacuation and dehydration.[1]

In Variable Refrigerant Flow (VRF) systems, the integrity of the internal chemical environment is entirely dependent on meticulous installation practices—specifically, the deep evacuation of the system using a two-stage vacuum pump. The industry standard dictates that a system must be evacuated to a sustained pressure of 500 microns (0.5 Torr) or lower, and hold that pressure, to ensure a dehydration state.[1] Failing to reach and hold this deep vacuum leaves behind residual non-condensable gases (primarily atmospheric nitrogen and oxygen) alongside liquid moisture. When a VRF system is commissioned and operates with a contaminated internal environment, a highly predictable, cascading chain of chemical and thermodynamic failures is inevitably initiated.[1] This report provides an exhaustive, peer-level technical analysis of the mechanisms triggered by improper vacuuming in R410A systems utilizing synthetic lubricants. It will explore the precise thermodynamic consequences of non-condensables via Dalton's Law, the complex chemical degradation pathways involving Polyolester (POE) lubricant hydrolysis and R410A thermal decomposition in Polyvinyl Ether (PVE) systems, the metallurgical phenomenon of galvanic copper plating, and the ultimate destruction of micrometric tolerances leading to mechanical and electrical scroll compressor burnout.

The Chemical Architecture of VRF Synthetic Lubricants

To accurately diagnose the chemical reactions within a contaminated VRF system, it is first necessary to classify the tribological agents at work. A common nomenclatural error in the field, reflected in the user query, is the interchangeable use of the terms Polyolester (POE) and Polyvinyl Ether (PVE). The query specifically references "FV50S POE (Polyolester) oil";

however, from a strict tribological and chemical standpoint, this is a misclassification. Daphne Hermetic Oil FV50S, manufactured by Idemitsu, is strictly a Polyvinyl Ether (PVE) base stock, not a Polyolester.[5]

Despite this classification distinction, the overarching mechanisms of moisture and air contamination severely impact both POE (via direct hydrolysis) and the broader PVE system (via refrigerant thermal degradation). Therefore, this analysis will correct the chemical classification of FV50S, detail the specific properties of PVE, and concurrently detail the exact hydrolysis mechanisms that affect POE oils, demonstrating how systemic burnout occurs regardless of the specific synthetic base stock employed.

Polyvinyl Ether (PVE) Architecture: The Daphne Hermetic FV50S

Polyvinyl Ether (PVE) is an innovative polymer-based synthetic lubricant developed primarily as an alternative to POE for HFC, HFO, and natural refrigeration systems.[8] The chemical backbone of PVE is characterized by an ether linkage (R–O–R'), which provides distinct tribological and chemical advantages over ester-based lubricants.[8] The fundamental advantage of the ether structure is its inherent immunity to hydrolysis.[5] Because the oxygen atom in the ether linkage is flanked by two alkyl or aryl groups without a highly reactive carbonyl carbon, the chemical bond is exceptionally stable and does not react with water to form acidic byproducts, even under elevated temperatures.[9]

Idemitsu's Daphne Hermetic Oil FV50S is an ISO-50 viscosity grade PVE formulated with specific anti-wear and antioxidant additives optimized for VRF and commercial refrigeration applications.[6] It exhibits excellent miscibility with R410A across a wide temperature spectrum, high dielectric strength necessary for inverter-driven compressors, and superior resistance to capillary tube blockage.[6]

Physical & Chemical Property	Daphne Hermetic FV50S Specification	Reference Test Method
Lubricant Classification	Polyvinyl Ether (PVE)	Idemitsu OEM Data [5]
Kinematic Viscosity Index	84	ASTM D2270 [5]
Density @ 20°C	0.9315 Kg/m³	ASTM D1298 [5]
Pour Point	-37.5 °C	ASTM D97 [5]
Flash Point (Cleveland Open Cup)	206 °C	ASTM D92 [5]

Total Acid Number (TAN)	< 0.01 mg KOH/g	ASTM D974 [5]
Moisture Content (Factory)	50 ppm	- [5]
Volumetric Resistivity (Dielectric)	$6 \times 10^{13} \Omega \cdot \mathrm{cm}$	- [6]

While FV50S is highly hygroscopic—meaning its polar nature will readily absorb ambient moisture from the atmosphere the moment it is exposed to air—it does not chemically degrade *via hydrolysis* when water is introduced.[3] However, the presence of water and air in a PVE system still serves as a catalyst for other severe systemic failures. While the oil itself may survive, the moisture drives the thermal decomposition of the R410A refrigerant itself, rendering the non-hydrolytic benefit of PVE moot in the face of improper evacuation.

Polyolester (POE) Architecture and Inherent Vulnerability

In contrast to the ether linkages of PVE, Polyolester (POE) is synthesized through a reversible esterification reaction. The formulation involves reacting a polyhydric alcohol (a polyol, typically containing a neopentyl carbon backbone such as pentaerythritol, dipentaerythritol, neopentyl glycol, or trimethylolpropane) with various linear or branched carboxylic acids.[13]

The chemical synthesis of POE is designed to provide excellent thermal stability, wide-ranging viscosity indexing, and superior boundary lubrication. The neopentyl backbone is sterically hindered, meaning the bulky molecular structure provides a degree of physical protection to the ester bonds. The specific carboxylic acids used in the synthesis dictate the final viscosity and refrigerant miscibility properties of the oil.[14] The formulation is meticulously optimized to ensure that the oil remains highly fluid at the low temperatures of the evaporator while simultaneously maintaining sufficient hydrodynamic film thickness at the extreme temperatures of the compressor discharge.[14]

However, the ester linkage (R–C(=O)–O–R') in POE represents the exact site of its fatal chemical vulnerability. Because the esterification reaction used to create the oil is fundamentally chemically reversible, the introduction of moisture drives the reaction backward.[15] When exposed to moisture under the extreme thermodynamic conditions of a running compressor, the ester bonds undergo hydrolysis, breaking the synthetic lubricant back down into its raw constituents: polyols and acids. Furthermore, because POE oil is constructed with polar molecular bonds to ensure HFC miscibility, it possesses a massive chemical affinity for water. A wet POE system can absorb approximately 2,500 parts per million (ppm) of moisture, which is an astronomical increase compared to legacy mineral oil, which has a maximum saturation limit of merely 25 ppm.[1]

The Physics of Inadequate Evacuation and Moisture Adsorption

The foundation of the systemic failure in VRF systems lies exclusively in the failure to reach a 500-micron vacuum. Atmospheric air is a mixture containing approximately 78% nitrogen, 21% oxygen, and varying amounts of water vapor depending on the relative humidity. When an installation technician fails to adequately evacuate the extensive copper line sets and indoor evaporator coils of a multi-zone VRF system, these elements remain permanently trapped inside the sealed architecture.

The Illusion of the Vacuum Plateau

A standard mechanical two-stage vacuum pump operates by continuously lowering the internal pressure of the system, which consequently lowers the vapor pressure and boiling point of any free water, allowing it to vaporize at ambient temperatures and be exhausted into the atmosphere.[1] However, vacuum pumps are governed by absolute physical limitations regarding chemical bonds. If a system utilizing POE or PVE oil has been exposed to the atmosphere for a prolonged period, the moisture does not merely sit on the copper surfaces as free liquid water; rather, it becomes chemically bonded to the polar molecules of the synthetic oil.[1]

Standard field evacuation alone cannot break this polar molecular bond. As a result, a technician pulling a vacuum on a moisture-saturated system will typically observe the digital micron gauge plateau and stall around 1,000 to 1,500 microns, regardless of how long the pump operates.[1] This plateau is a critical diagnostic indicator: the free moisture on the copper walls has been successfully vaporized and removed, but the chemically bound moisture remains deeply saturated within the synthetic lubricant. To physically remove this bound moisture, the system relies entirely on physical adsorption via high-capacity liquid-line filter driers containing desiccant cores (such as specialized molecular sieves or activated alumina).[1] If the filter drier's capacity is exceeded by the sheer volume of trapped water, or if the filter drier is bypassed, the moisture remains active and highly destructive within the circuit.

Dalton’s Law of Partial Pressures and Thermodynamic Strain

While moisture degrades the chemistry, the trapped atmospheric air (primarily nitrogen and oxygen) severely degrades the thermodynamics. When air and nitrogen are left in the system, they are deemed "non-condensables." Unlike the R410A refrigerant, which readily shifts between liquid and vapor phases depending on pressure and temperature, these gases do not condense into a liquid in the condenser coil at normal HVAC operating pressures.[17] Instead, these non-condensable gases remain permanently in a vapor state, occupying critical physical volume at the top of the condenser coil and receiver tank, thereby reducing the effective surface area available for refrigerant heat rejection.

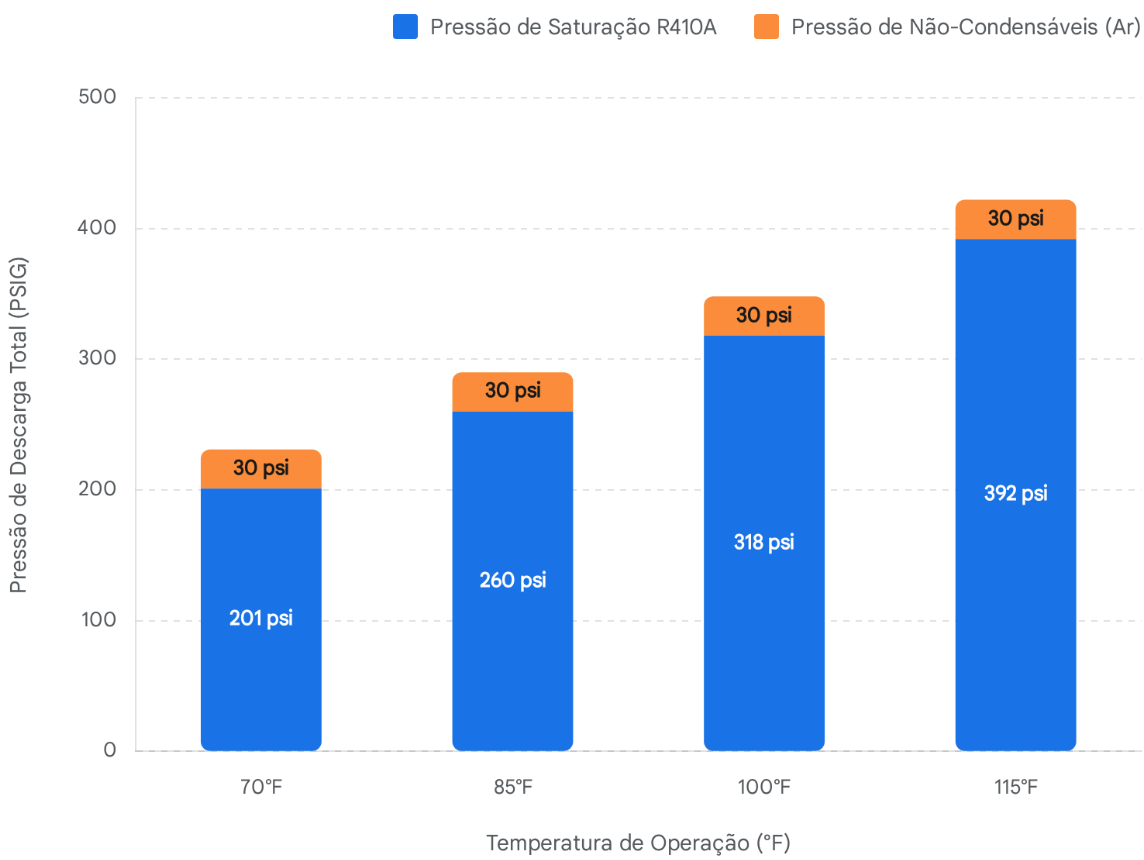
The total internal pressure within this closed VRF system is strictly governed by Dalton’s Law of Partial Pressures. This thermodynamic law states that in a mixture of non-reacting gases, the total pressure exerted on the container walls is equal to the sum of the partial pressures of the individual gases present.[18]

Mathematically, this is expressed as:

$$P_{\text{total}} = P_{\text{refrigerant}} + P_{\text{air}} + P_{\text{nitrogen}}$$

In a normal, properly evacuated VRF system operating in cooling mode on a 95°F day, the expected high-side (discharge) pressure for R410A typically fluctuates between 400 and 450 psi.[22] However, if non-condensables are present, their partial pressure is added directly to the refrigerant's saturation pressure. This inescapable physical reality forces the high-side pressure to spike significantly above normal operational parameters.[18]

Impacto da Contaminação por Não-Condensáveis nas Pressões do Sistema



O ar atmosférico não-condensável retido no sistema atua de forma independente do fluido frigorigéneo. Seguindo a Lei de Dalton, a pressão do ar retido adiciona-se diretamente à pressão de saturação do R410A, forçando o compressor a operar contra pressões de descarga anormalmente altas.

Fontes de dados: Refrigerants Center, BPA, NRC, SunlitBD, Reddit r/HVAC

Compression Ratio Alteration and Extreme Discharge Temperatures

The artificial elevation of the discharge pressure drastically alters the system's compression ratio ($CR = P_{\text{discharge}} / P_{\text{suction}}$). Because the suction pressure remains relatively constant (dictated by the evaporator load), the spike in discharge pressure significantly increases the CR. The scroll compressor must therefore expend exponentially more mechanical and electrical energy to compress the refrigerant vapor against the artificially high pressure created by the trapped atmospheric air.[18]

This extreme mechanical and thermodynamic strain directly translates to immense heat generation. The relationship between pressure and temperature during the compression stroke of the scroll compressor can be approximated by the isentropic compression equation for ideal gases:

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

Where:

- T_2 is the absolute discharge temperature.
- T_1 is the absolute suction temperature.
- P_2 is the absolute discharge pressure.
- P_1 is the absolute suction pressure.
- k is the specific heat ratio of the gas mixture.

As P_2 increases due to the partial pressure of the non-condensables, T_2 rises exponentially.

Furthermore, the presence of air inherently alters the overall specific heat ratio (k) of the gas mixture being compressed, further compounding the temperature rise across the compression stroke. Consequently, the compressor discharge line temperatures easily exceed safe operational limits (typically designed around 225°F for modern scrolls).[21] These extreme thermal conditions, combined with the presence of reactive oxygen and moisture, serve as the perfect catalytic crucible for the chemical destruction of the entire system.

Catalysis and Chemical Degradation Pathways

With the system operating under excessive heat, abnormal high-side pressures, and carrying both moisture and oxygen, profound chemical degradation begins. The exact chemical pathway depends entirely on the type of synthetic oil used—POE or PVE—but both pathways inevitably lead to the massive generation of highly corrosive acids, fulfilling the next stage of the pathology.

Pathway A: The Autocatalytic Hydrolysis of POE Oil

If the VRF system relies on Polyolester (POE) oil, the primary and most devastating degradation mechanism is hydrolysis.[1] The extreme heat of the compressor discharge line, combined with the continuous, violent turbulent mixing of the lubricating oil and refrigerant, forces the bound water molecules to react directly with the ester linkages of the POE polymer chain.
The generalized chemical equation for POE hydrolysis is:

$$\text{Polyol Ester (POE)} + \mathrm{H_2O} \rightleftharpoons \text{Polyhydric Alcohol} + \text{ Carboxylic Acids}$$

For example, if the POE is based on a standard pentaerythritol backbone, the hydrolysis reaction sequentially cleaves the ester branches, breaking the complex lubricant down into free pentaerythritol and free linear or branched carboxylic acids.[16]
The kinetics of this specific reaction are particularly devastating to closed-loop refrigeration systems because ester hydrolysis is an **autocatalytic reaction**.[1] This means that the carboxylic acid produced by the initial reaction acts as an active catalyst that violently accelerates the rate of further hydrolysis.[1]
The reaction rate can be mathematically modeled using a third-order kinetic formula. Assuming the reverse reaction (esterification) is negligible under active breakdown conditions, the formation of acid over time (t) follows a complex sigmoidal (S-shaped) curve defined by the following kinetic equation [24]:

$$[\text{Acid}]_t = \frac{{}_0 + [\text{Acid}]_0}{1 + \left(\frac{{}_0}{[\text{Acid}]_0}\right) \times \exp\left(-\left({}_0 + [\text{Acid}]_0\right) + [\text{POE}]_0 \times k \times t\right)}$$

Where:

- k is the evaluated third-order reaction rate constant.
- ${}_0$ is the initial concentration of retained water.
- $[\text{Acid}]_0$ is the initial concentration of acid (acting as the active catalyst).

In a new, pristine system with proper evacuation, $[\text{Acid}]_0$ is near zero, and the uncatalyzed reaction is exceedingly slow (often calculated to be at least 1,000 times slower than the acid-catalyzed rate).[24] However, once a minute amount of acid is generated by the extreme temperatures caused by the non-condensables, the reaction rate k increases exponentially.
Thermodynamic evaluations of pentaerythritol esters with 1,000 mg/kg of water at 150°C reveal that linear acids (like Pentaerythritol-tetra heptanoate) possess a rate constant of $1.71 \times 10^{-5}\ \text{l/mol-sec}$ and an activation enthalpy ($\Delta H_{\text{activation}}$) of 11.8 kcal/mole.[24]

Alpha-branched acids (like Pentaerythritol-tetra 2-ethyl-hexanoate) offer slightly more steric hindrance, slowing the rate constant to $2.89 \times 10^{-6}\ \mathrm{l/mol\text{-}sec}$, but ultimately succumbing to the heat.[24]

As the activation enthalpy is rapidly overcome by the compressor's abnormally elevated discharge heat, more acid forms, which in turn accelerates the creation of even more acid in a runaway, inescapable feedback loop. This drives the Total Acid Number (TAN) of the lubricant far beyond safe operational parameters, destroying its lubricity and turning it into a corrosive solvent.[2]

Mitigation Attempts via Epoxide Acid Catchers

To combat this known vulnerability, manufacturers sometimes formulate POE with "Acid Catchers" (AC), typically epoxide-based compounds like neodecanoyl glycidyl ester. These additives improve hydrolytic stability by chemically reacting with and trapping free acids as they form, preventing the autocatalytic loop.[24] The thermodynamics of the 3-atom epoxide ring opening are highly favorable, with an acid-trapping reaction enthalpy (ΔH) of -30.1 kcal/mole.[24] However, at the extreme temperatures generated by high non-condensable pressures, these finite additives are rapidly depleted, and the system inevitably succumbs to runaway hydrolysis.

Pathway B: Thermal Pyrolysis of R410A Refrigerant in PVE Systems

If the system utilizes Idemitsu Daphne Hermetic Oil FV50S (a Polyvinyl Ether), the oil itself structurally resists hydrolysis.[5] However, the system is absolutely not safe from catastrophic acid generation. The extreme heat caused by the non-condensables forces a different, equally destructive chemical reaction: the thermal decomposition (pyrolysis) of the R410A refrigerant.[28]

R410A is a near-azeotropic mixture composed of exactly 50% R32 (difluoromethane, CH_2F_2) and 50% R125 (pentafluoroethane, CHF_2CF_3) by weight.[29] Under normal, properly evacuated operating conditions, these molecules are highly stable. However, when exposed to localized "hot spots" exceeding 250°C (482°F) inside the compressor—temperatures easily and frequently reached when internal friction increases and non-condensable head pressures spike—the strong carbon-fluorine molecular bonds of the hydrofluorocarbons begin to violently fracture.[28]

The presence of trapped atmospheric oxygen and moisture reacts violently with the fracturing refrigerant molecules. The decomposition products of R410A under these extreme thermodynamic conditions include highly toxic and incredibly corrosive compounds:

1. **Hydrogen Fluoride (HF)**: A highly reactive gas that immediately dissolves in available system moisture to form extremely corrosive hydrofluoric acid.[28]
2. **Carbonyl Fluoride (COF_2)**: A short-lived, extremely toxic intermediate compound that rapidly and heterogeneously hydrolyzes in the presence of water to produce carbon

dioxide and additional hydrofluoric acid ($COF_2 + H_2O \rightarrow CO_2 + 2HF$).[28]

The Convergence of Failures: A Dual Cascade

The failure mechanism represents a dual cascade of chemical and mechanical deterioration depending on the chosen chemistry. Inadequate evacuation leaves both moisture and air in the system, driving severe thermodynamic stress via Dalton's Law. If POE is utilized, the heat and moisture drive autocatalytic hydrolysis, yielding vast quantities of carboxylic acids. Conversely, if PVE is utilized, the oil survives, but the excessive heat triggers R410A thermal breakdown, yielding hydrofluoric acid (HF). Despite these disparate origins, both chemical pathways ultimately converge on the exact same catastrophic outcome: intense heavy metal corrosion of the copper line sets, transport of these metallic salts to the compressor, galvanic displacement (copper plating), and eventual mechanical scroll seizure. Thus, the system is doomed regardless of whether the oil itself is hydrolytically stable.

The Metallurgy and Electrochemistry of Copper Plating

The generation of either carboxylic acid (via POE) or hydrofluoric acid (via R410A) fundamentally and permanently alters the pH of the entire system. The extensive internal network of a VRF system is composed almost entirely of copper tubing (suction lines, liquid lines, branch controllers, and indoor evaporator coil networks). When exposed to strong, hot acids, these internal copper surfaces begin to undergo rapid chemical corrosion.

Formicary Corrosion and Copper Dissolution

The highly acidic internal environment actively strips the natural, protective oxide layer from the internal copper surfaces.[1] The acids aggressively attack the bare copper tubing, creating microscopic, branching tunnel networks into the metal in a process known as formicary (or "ant's nest") corrosion.[1]

During this active corrosion process, solid elemental copper (Cu) is oxidized and dissolved into the circulating mixture of liquid refrigerant and synthetic oil as cupric (Cu^{2+}) or cuprous (Cu^{+}) ions.[2] Depending on the acid present, these ions form organic or inorganic copper salts, such as copper carboxylate or copper fluoride. The continuous high-velocity flow of the liquid and vapor refrigerant acts as an efficient transport mechanism, carrying these suspended heavy metal copper salts out of the line sets and directly into the heart of the system: the compressor.[36]

The Galvanic Displacement Reaction

As the dissolved copper ions enter the hot compressor shell, they encounter the internal mechanical components, which are primarily manufactured from cast iron and forged steel alloys. These components include the crankshaft, the main and sub-bearings, the Oldham

coupling, and the involute scroll wraps.[36]
Due to the extreme heat generated by the non-condensables (often acting directly on localized boundary friction points such as the bearings or discharge valves), the thermodynamic conditions become optimal for a spontaneous, irreversible electrochemical reaction known as galvanic displacement (or immersion plating).[40]
Galvanic displacement is an oxidation-reduction (redox) reaction driven strictly by the difference in standard electrode potentials between two different metals in an electrolytic solution (in this case, the highly acidic oil).[44] When a more noble metal ion (such as dissolved copper) comes into contact with a less noble, more anodic solid metal (such as iron or steel), the ions in the solution will spontaneously strip electrons from the solid metal to stabilize themselves.
The standard reduction potentials (E° versus Standard Hydrogen Electrode at 25°C) precisely dictate the direction and spontaneity of the reaction:

Electrochemical Half-Reaction	**Standard Potential (E◦, Volts)**	**Metal Characterization**
$\mathrm{Cu}^{2+} + 2e^{-} \rightarrow \mathrm{Cu}_{(s)}$	+0.34 V	More Noble (Cathodic)
$\mathrm{Fe}^{2+} + 2e^{-} \rightarrow \mathrm{Fe}_{(s)}$	-0.44 V	Less Noble (Anodic)

Data derived from Standard Potential Tables.[41]
Because the standard potential of copper is significantly higher (more positive) than that of iron, the overall displacement reaction is thermodynamically favorable, meaning the Gibbs Free Energy (ΔG) is negative, and the reaction proceeds spontaneously without external current:

$$\mathrm{Fe}_{(s)} + \mathrm{Cu}^{2+}_{(aq)} \rightarrow \mathrm{Fe}^{2+}_{(aq)} + \mathrm{Cu}_{(s)}$$

In this redox reaction, the iron atoms on the highly machined surface of the compressor's bearings and scroll flanks act as sacrificial anodes. They lose two electrons (oxidize) and dissolve into the contaminated oil as ferrous (Fe^{2+}) ions, essentially pitting and degrading the finely machined steel surfaces.[41] Simultaneously, the dissolved copper ions arriving from the line sets accept these electrons (reduce) and precipitate directly onto the steel surface as a solid, elemental copper film.[35]
This is the phenomenon universally referred to in the industry as **copper plating**. It is critical to understand that this is not merely a deposition of loose particulate dust; it is a chemically bonded, continuously growing metallic layer of copper physically replacing the surface steel of the hottest, most critical moving parts of the compressor.[36]

Mechanical Degradation and Scroll Compressor Burnout

The internal architecture of a modern VRF scroll compressor represents a marvel of precision engineering. The efficiency of the continuous compression cycle relies entirely on maintaining exact, microscopic clearances between the high-speed moving parts. The chemical and metallurgical degradation detailed above violently disrupts this delicate balance, guaranteeing a catastrophic mechanical and electrical failure.

Loss of Lubricity and Viscosity Breakdown

Long before the galvanic copper plating reaches a critical thickness, the primary defense against mechanical wear—the hydrodynamic oil film—is compromised.[1] As the POE oil undergoes severe hydrolysis, or as the PVE oil becomes heavily saturated with hydrofluoric acid and dissolved ferrous iron sludge, its kinematic viscosity and boundary lubricity characteristics are permanently altered.[36]

The Total Acid Number (TAN) rises rapidly, and the mixture of dissolved metals, carboxylic or hydrofluoric acid, and degraded polymer chains agglomerates into a thick, abrasive, highly viscous sludge.[36] This sludge restricts the internal, capillary-sized oil channels of the crankshaft, severely starving the upper thrust bearings and the orbiting scroll of vital hydrodynamic lubrication.[37] As the oil film thins and fails, the system transitions from a safe, low-friction hydrodynamic lubrication regime into a destructive boundary lubrication regime, where direct metal-to-metal contact begins to occur, generating even more localized heat and accelerating the chemical breakdown.

Micron-Level Tolerance Destruction

Scroll compressors achieve gas compression by trapping vapor in crescent-shaped pockets formed by the precise meshing of two involute spirals: one stationary (the fixed scroll) and one orbiting (the orbiting scroll).[48] To achieve high volumetric efficiency without using physical polymer seals, the compressor relies on ultra-precise radial and axial compliance, utilizing a micron-thin film of oil to seal the gaps between the high-pressure and low-pressure zones.[48]

The flanks of the scroll wraps are machined to exacting tolerances, frequently less than 20 micrometers (0.00078 inches), with surface finishes engineered in the single-micron range.[49] Similarly, the journal bearings and the main shaft must maintain specific, tight clearance ratios to allow for the hydrodynamic oil film to develop and support the extreme radial loads of the compression cycle.[51] Research indicates that even a marginal bearing clearance increase to just 0.1 mm can cause severe oscillation, dynamic shock, and massive vibration of the orbiting scroll.[51]

As the galvanic displacement reaction proceeds unchecked, the elemental copper plating continuously thickens on these critical bearing surfaces, the Oldham coupling keys, and the discharge valve areas.[36] Copper is a highly ductile metal. As it plates onto the journal bearings and the flanks of the orbiting scroll, it systematically bridges the carefully engineered micro-clearances.[37] The compressor physically runs out of room to rotate.

Mechanical Seizure and Electrical Dielectric Collapse

With the operational clearances reduced to zero by the intrusive copper layer and the protective oil film destroyed by acid sludge, the coefficient of friction inside the compressor skyrockets.[37] The severe friction generates immense localized heat, causing rapid thermal expansion of the orbiting scroll, the fixed scroll, and the main shaft.

Eventually, the copper-plated, thermally expanding components exceed the available geometric tolerances of the compressor housing. The bearing surfaces and scroll flanks physically lock together in a process known as galling or cold welding.[47] The orbiting scroll violently seizes against the fixed scroll, or the main shaft seizes within its copper-plated journal bearing. The mechanical rotation of the system comes to an immediate, violent halt.[38]

At the exact fraction of a second of mechanical seizure, the compressor motor enters a catastrophic "locked rotor" condition. The inverter drive or contactor continues to supply voltage, and the motor stator continues to draw maximum inrush current (Locked Rotor Amps, or LRA) in a futile, continuous attempt to overcome the immense, insurmountable mechanical resistance.[47] This massive electrical draw generates extreme inductive heat within the copper motor windings.

Simultaneously, the dielectric strength (volumetric resistivity) of the contaminated PVE or POE oil—originally engineered to be highly resistive at $6 \times 10^{13} \Omega \cdot \mathrm{cm}$ [6]—has been severely compromised by the high concentration of conductive water, highly ionized acid, and suspended copper/iron metallic particulates.[8] The extreme heat of the locked rotor melts the insulating enamel coating on the copper motor windings. With the physical insulation destroyed and the entire motor bathed in a highly conductive, acidic fluid, the high-voltage electricity shorts directly across the phases of the windings or jumps to the steel compressor shell in a massive ground fault.[47] This massive electrical arc instantly vaporizes the surrounding contaminated refrigerant and oil, completing the systemic burnout, carbonizing the internals, and permanently destroying the compressor.

The failure of a VRF system due to improper evacuation is not merely a random mechanical anomaly; it is the inevitable conclusion of a rigorous, unforgiving thermodynamic and chemical sequence. The failure to achieve a deep 500-micron vacuum allows non-condensable atmospheric gases to remain trapped within the architecture, artificially inflating discharge pressures via Dalton's Law and exponentially increasing the thermal load on the compressor. This extreme heat serves as the ultimate catalyst for the system's destruction. If the system employs Polyolester (POE) oil, the retained moisture drives an autocatalytic hydrolysis reaction, rapidly cleaving the ester bonds into highly corrosive carboxylic acids. If the system correctly utilizes a Polyvinyl Ether (PVE) oil, the oil resists hydrolysis, but the excessive discharge heat violently decomposes the R410A refrigerant itself, yielding highly toxic hydrofluoric acid. Regardless of the synthetic base stock, the resultant acidic environment viciously attacks the system's internal copper line sets. Through a galvanic displacement reaction, the dissolved copper ions are transported to the hot steel surfaces of the compressor bearings and scroll wraps, where they displace the sacrificial iron and plate as solid elemental copper. By physically obliterating the micron-level clearances required for scroll compression and simultaneously destroying the hydrodynamic oil film, the copper plating induces overwhelming mechanical friction. The final event is an inescapable mechanical seizure of the orbiting components, triggering a massive electrical overload, the melting of motor winding insulation, and a catastrophic electrical burnout. Meticulous evacuation practices are, therefore, not merely a best practice recommendation, but the absolute fundamental physical prerequisite for preserving the chemical equilibrium and operational lifespan of modern synthetic refrigeration systems.

Referências citadas

1. POE Oil and Moisture: Why Evacuation Alone Fails on Modern ..., acessado em junho 2, 2026, https://hvacknowitall.com/blog/poe-oil-and-moisture-why-evacuation-alone-fails-on-modern-hvac-systems
2. arti-21cr/611-50060-01 using acid number as a leading indicator of refrigeration and air conditioning system - OSTI.GOV, acessado em junho 2, 2026, https://www.osti.gov/servlets/purl/823889
3. Best Practices for HFO Refrigerants | PDF - Scribd, acessado em junho 2, 2026, https://www.scribd.com/document/679986629/hvacrschool-com-Oil-Best-Practices-for-HFC-and-HFO-based-Systems
4. Polyol ester (POE) base refrigeration lubricant, acessado em junho 2, 2026, https://www.klsummit.com/en/about-us/newsroom/news/is-your-lubricant-fit-to-withstand-the-long-term-challenges-inherent-with-carbon-dioxide-co2-refrigeration-system-operation/
5. Idemitsu PVE lubricant - Daphne Hermetic Oil FV50S - 5 litres metal can - General Gas, acessado em junho 2, 2026, https://www.generalgas.eu/scheda-prodotto/lubrificante-pve-idemitsu-daphne-hermetic-oil-fv50s-tanica-in-metallo-da-5-litri.html
6. Daphne Refrigeration Oil FV50S (5L) - Reciprocating Compressor, acessado em junho 2, 2026, https://qishanr.com/products/daphne-refrigeration-oil-fv50s-5l
7. Idemitsu PVE Lubricants - Nu-Calgon, acessado em junho 2, 2026, https://www.nucalgon.com/products/refrigeration-vp-oils/idemitsu-pve-lubricants/
8. Advantages of PVE Refrigerant Oil | PDF | Applied And Interdisciplinary Physics - Scribd, acessado em junho 2, 2026, https://www.scribd.com/document/270099491/PVE-lube-oil-EN-pdf
9. CATALOGUE RACHP, acessado em junho 2, 2026, https://www.generalgas.it/storage/catalogo/generalgas-catalogue-RACHP-2025-10.pdf
10. Nu-Calgon 4319-20 - Refrigeration Oil, PVE-FV50S 6 x 1 qt. cs. - Beenimax, acessado em junho 2, 2026, https://www.beenimax.com/en/nuc-4319-20.html
11. Idemitsu PVE lubricant - Daphne Hermetic Oil FV50S - 1 litre metal can - General Gas, acessado em junho 2, 2026, https://www.generalgas.eu/scheda-prodotto/lubrificante-pve-idemitsu-daphne-hermetic-oil-fv50s-tanica-in-metallo-da-1-litro.html

12. pve-fv50s refrigeration oil, 6x1 qt cs - Nu-Calgon Product Bulletin, acessado em junho 2, 2026, https://www.nucalgon.com/media/8928/3-88_pve_oil-product-bulletin.pdf
13. US9587202B2 - Process for reducing the total acidity of refrigerating compositions - Google Patents, acessado em junho 2, 2026, https://patents.google.com/patent/US9587202B2/en
14. US9951289B2 - Refrigeration oil - Google Patents, acessado em junho 2, 2026, https://patents.google.com/patent/US9951289B2/en
15. REFRIGERATION, acessado em junho 2, 2026, https://azaranstore.com/wp-content/uploads/2022/09/271-ASHRAE-HANDBOOK-REFRIGERATION-2014.pdf
16. Refrigerant System Chemistry Overview | PDF | Chlorofluorocarbon | Greenhouse Gas - Scribd, acessado em junho 2, 2026, https://www.scribd.com/document/565085141/R05S
17. bulletin 130 - Shipco® Pumps, acessado em junho 2, 2026, https://shipcopumps.com/content/support/download.php?id=bulletin-130
18. 7 Essential Tips For HVAC Refrigerant Charging And System Evacuation - Sunlit BD, acessado em junho 2, 2026, https://sunlitbd.com/hvac-refrigerant-charging-and-system-evacuation/
19. method of characteristics for column closure waterhammer - Nuclear Regulatory Commission, acessado em junho 2, 2026, https://www.nrc.gov/docs/ML0217/ML021720773.pdf
20. Design Load Components for Vacuum Systems - Welcome to Thermo Systems Inc., acessado em junho 2, 2026, https://www.thermosysinc.com/inquiry1.html
21. How do you diagnose non condensables? : r/HVAC - Reddit, acessado em junho 2, 2026, https://www.reddit.com/r/HVAC/comments/qs0r6u/how_do_you_diagnose_non_condensables/
22. R410A Operating Pressures | A Complete Guide - Royal Refrigerants, acessado em junho 2, 2026, https://royalrefrigerants.com/blogs/news/r410a-operating-pressures-a-complete-guide
23. R410A Operating Pressures: Charts, Readings, and Best Practices - Refrigerants Center, acessado em junho 2, 2026, https://refrigerantscenter.com/blogs/refrigerant-review/r410a-operating-pressures-charts-readings-and-best-practices
24. Improving Hydrolytic Stability of POE Lubricants by ... - Purdue e-Pubs, acessado em junho 2, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1488&context=iracc
25. Experimental Investigation on the Effects of POE Oil and Iron Powder on the Corrosion of TP2 Copper Tubes in Acetic Acid Vapors - MDPI, acessado em junho 2, 2026, https://www.mdpi.com/2076-3417/15/16/9224
26. Biodegradable Polymers for Microencapsulation of Drugs - PMC - NIH, acessado em junho 2, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6147704/
27. Thermal Decomposition Kinetics of Polyol Ester Lubricants | Energy & Fuels,

acessado em junho 2, 2026, https://pubs.acs.org/doi/10.1021/acs.energyfuels.6b01863
28. R-410A SDS / MSDS: Safety Data Sheet Reference (Plus PDF Download) - AC Direct, acessado em junho 2, 2026, https://www.acdirect.com/blog/r410a-sds-msds-safety-data/
29. R-410A Safety Data Sheet - National Refrigerants, Inc., acessado em junho 2, 2026, https://refrigerants.com/wp-content/uploads/2019/12/SDS-R410A.pdf
30. R410A Safety Data Sheet Overview | PDF | Combustion | Fires - Scribd, acessado em junho 2, 2026, https://www.scribd.com/document/971096505/R410A-SDS
31. Safety data sheet - Daikin Chemicals, acessado em junho 2, 2026, https://www.daikinchemicals.com/library/pb_common/pdf/sds/Refrigerants/sds-R-410A-E_20230323.pdf
32. Safety Data Sheet according to Regulation (EC) No. 1907/2006 (REACH) - Rittal, acessado em junho 2, 2026, https://www.rittal.com/imf/none/5_1000/Rittal_Safety_data_sheet_Refrigerant_R410A_5_1000/
33. Refrigeration, Air Conditioning and Heat Pumps Technical Options Committee - Ozone Secretariat, acessado em junho 2, 2026, https://ozone.unep.org/system/files/documents/RTOC-assessment%20-report-2022.pdf
34. R410A ! SECTION 1: Identification of the substance/mixture and of the company/undertaking ! SECTION 2: Hazards identification, acessado em junho 2, 2026, https://www.brgroup.eu/site/files/R410A%20BRG.pdf
35. (PDF) Study on the coating performance of copper layer - ResearchGate, acessado em junho 2, 2026, https://www.researchgate.net/publication/398882665_Study_on_the_coating_performance_of_copper_layer
36. Chemical Aspects of Refrigerant Systems - Purdue e-Pubs, acessado em junho 2, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1074&context=icec
37. Compressor Lubrication, acessado em junho 2, 2026, https://sanden.com/wp-content/uploads/2024/05/MACS-2020-Sanden-Presentation-Lube-MR4.pdf
38. Practical tips on AC service and compressor replacement - denso-am.eu, acessado em junho 2, 2026, https://assets.denso-am.eu/production/attachments/DENSO-AC_brochure_tips-and-tricks_EN.pdf
39. A/C COMPRESSOR GUIDE AND PRACTICAL TIPS - Sanden USA, acessado em junho 2, 2026, https://sanden.com/objects/AC_compressor_guide_USA_20210730.pdf
40. Articles - Jarnistech, acessado em junho 2, 2026, https://www.jarnistech.com/articles
41. Assessment Of Industrial Hazardous Waste Practices Electroplating And Metal Finishing Industries, Job Shops - epa nepis, acessado em junho 2, 2026, https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=2000Y3DA.TXT
42. Copper Nickel Chromium and Zinc Segment of the Electroplating Point Source

Category. - epa nepis, acessado em junho 2, 2026, https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=10004HX2.TXT
43. What causes this copper plating on compressor valves? : r/HVAC - Reddit, acessado em junho 2, 2026, https://www.reddit.com/r/HVAC/comments/a2g61j/what_causes_this_copper_plating_on_compressor/
44. ASM, acessado em junho 2, 2026, https://download.s21i.faiusr.com/11753481/0/2/ABUIABA9GAAgy-aHlwYotcyZ4gc.pdf?f=13%E3%80%81ASM%20Metals%20Handbook%2C%20Vol%2013.pdf&v=1658977101
45. Corrosion: Fundamentals, Testing, and Protection ASM - UFPR, acessado em junho 2, 2026, http://ftp.demec.ufpr.br/disciplinas/TM315/Asm%20Metals%20Handbook%20Volume%2013%20-%20Corrosion%20Fundamentals,%20Testing,%20And%20Protection.pdf
46. Electrodeposition and Surface Finishing - National Academic Digital Library of Ethiopia, acessado em junho 2, 2026, http://ndl.ethernet.edu.et/bitstream/123456789/70954/1/2014_Book_ElectrodepositionAndSurfaceFin.pdf
47. Ever Hear of Copper Plating Inside the Compressor? - HVAC School, acessado em junho 2, 2026, http://www.hvacrschool.com/ever-hear-of-copper-plating-inside-the-compressor/
48. SCROLL COMPRESSORS - Carrier, acessado em junho 2, 2026, https://www.shareddocs.com/hvac/docs/1001/Public/02/811-20065.pdf
49. Optimum processing through component and market expertise: solutions for the scroll compressor | MAPAL, acessado em junho 2, 2026, https://mapal.com/en-us/a/process%20solutions-scroll%20compression
50. A Study of Designing 35RT Aluminum Alloy Scroll Compressor - Purdue e-Pubs, acessado em junho 2, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2741&context=icec
51. Simulation research on dynamic characteristics of scroll compressor with flexible clearance, acessado em junho 2, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11790136/
52. Journal Bearing Performance in a Scroll Compressor - Purdue e-Pubs, acessado em junho 2, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1880&context=icec
53. Characteristics of the Plain Bearing in Scroll Compressors - Purdue e-Pubs, acessado em junho 2, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1843&context=icec
54. US6439867B1 - Scroll compressor having a clearance for the oldham coupling - Google Patents, acessado em junho 2, 2026, https://patents.google.com/patent/US6439867B1/en